Research Suite - Complete Map

LEONARDO V2

Scope, architecture, workflows, object lifecycles and Leonardo integration

A visual companion to the complete documentation
Use this map to understand the system and follow its main paths. Keep the four Research manuals beside it for field-level contracts, algorithms, complete failure rules and implementation detail.

Research is Leonardo's interactive historical-chart and Financial Tools workspace. It consumes accepted data, coordinates explicit calculations and saves reusable research context through existing canonical services.

Document basis	Current reference
Edition / prepared	1.0 / 8 October 2026
Working candidate	main at deb24138042d2de524f38d0b230c1aad7611f83e, plus accepted uncommitted corrections and documentation.
Environment verified	Python 3.12.12 / PySide6 6.10.2
Code validation already recorded	2,831 full-suite passes / 7 skips; later independent audit: 115 offscreen checks and 21 native checks.
Documentation relationship	Additive overview. No manual is replaced, shortened or made obsolete.
Format reference	Leonardo V2 Core - Complete Map, 16 July 2026. Structure and visual language only; current RS facts come from current sources.

The current candidate includes the 16-chart extension and final Notebook transition corrections. Historical ZIPs and the July Core report are not the current Research baseline.

Reading guide and navigation

Read overview and specification together
Start here for orientation; use the linked entry manual for complete behavior. The PDF has bookmarks and a clickable contents list. Section references below point to the unchanged detailed manuals.

Reference	Document in this same folder
S - Scope and behavior	RESEARCH_SUITE.md
A - Architecture	RESEARCH_ARCHITECTURE.md
L - Workflows and lifecycles	RESEARCH_LIFECYCLES.md
D - Data and interfaces	RESEARCH_DATA_CONTRACTS.md

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Diagram key: blue = Research or GUI state; green = accepted/durable/domain outcome; amber = decision or guarded boundary; red = refusal/failure. Arrows show the stated call, data or lifecycle direction, not new authorities.

Purpose, scope and system context

Research lets a user inspect accepted historical markets, compare independent charts, calculate Studies, save reusable results and definitions, and retain observations. Locally accepted data can be used without a live provider connection.

Governing rule
Research coordinates research work. Core owns execution; OHLCV owns accepted source truth; Financial Tools owns mathematics; canonical stores own saved objects.

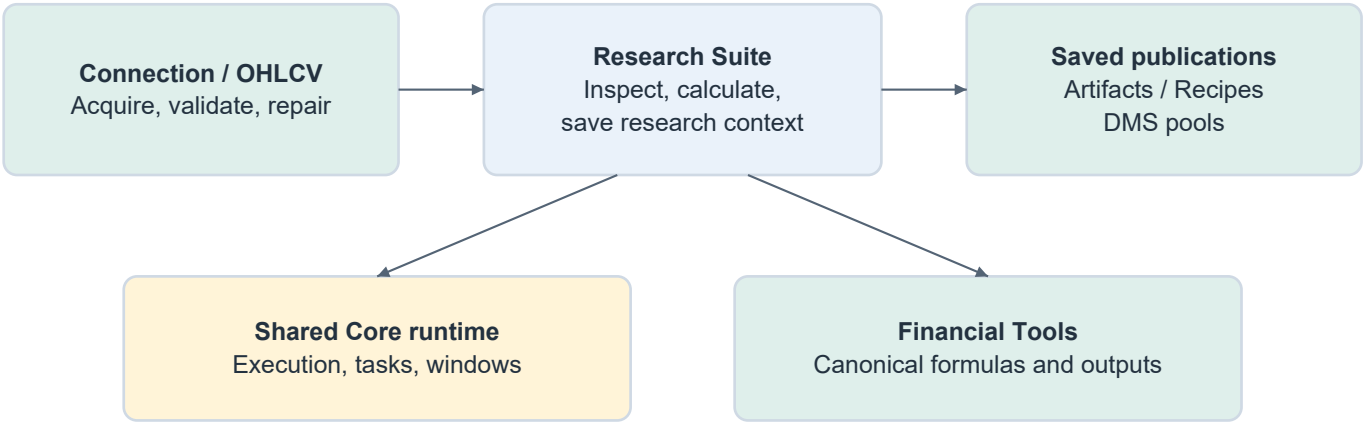


Figure 1. Research consumes accepted OHLCV, calls shared authorities, and explicitly publishes saved results. Research-specific pools preserve context separately.

Inside the Research workflow	Outside its authority
Charts, Studies, views, Environment application	Provider communication, download, repair and validation decisions
Environment, Snapshot and Notebook context	Financial formulas or a second Artifact/Recipe identity policy
Explicit Apply / Edit / Save and GUI progress	Database construction, reviewed imports, revisions and append policy
Potential Trades, Notes and Points of Interest	Order execution, risk approval, backtesting or Analysis engines

Six concepts that must not collapse into one

Dataset: immutable accepted candles. **Study:** one chart's transient result. **Artifact:** saved calculation. **Environment:** ordered Study definitions. **Snapshot:** chart definitions and workspace state. **Notebook:** independent observations, optionally referenced by Snapshots.

Active production composition

Leonardo constructs shared services first, then the GUI. A Suite receives those existing services; opening a window does not establish new stores, a private runtime or another task owner.

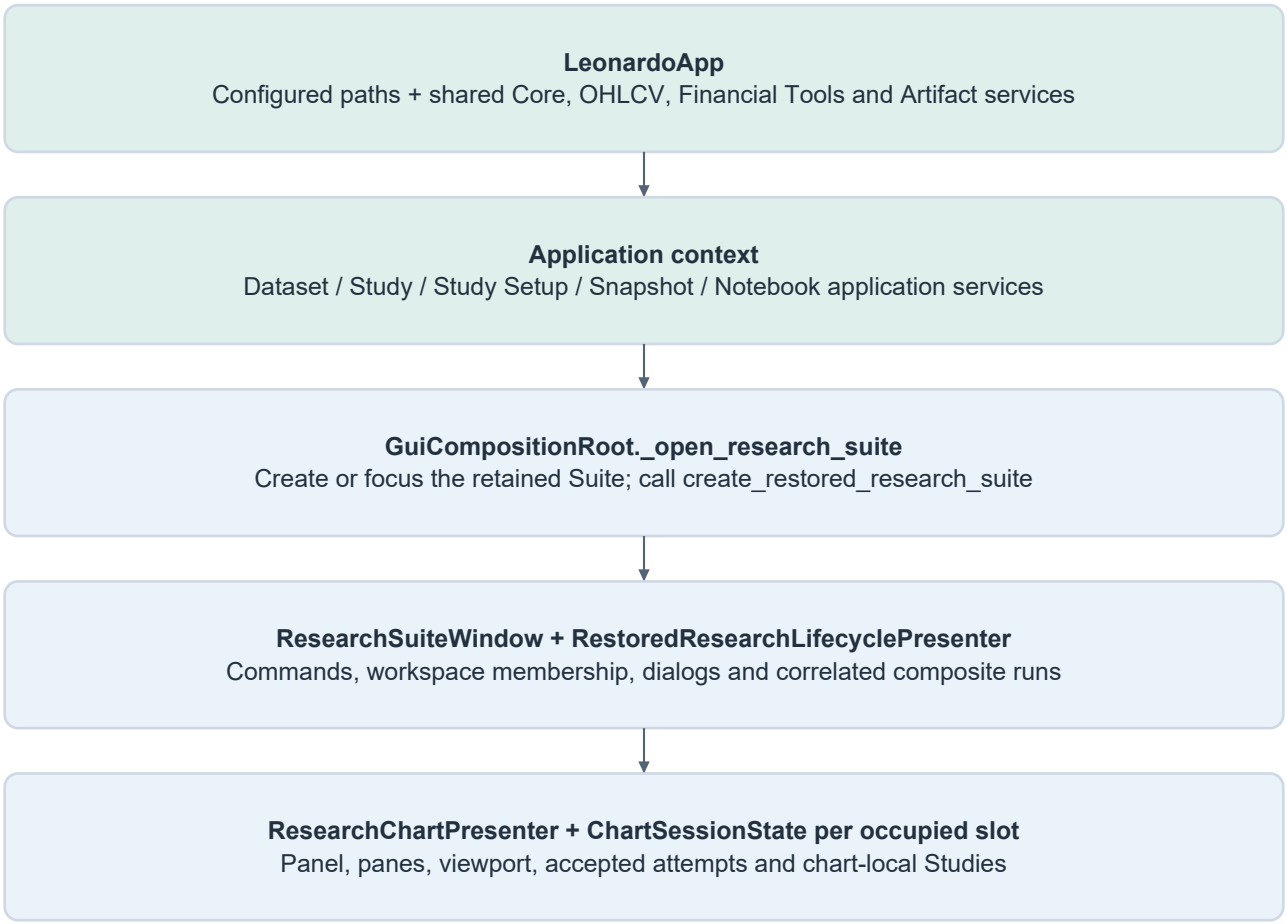


Figure 2. Construction and GUI ownership. Long operations return through application services and Core; this hierarchy does not place domain calculations inside widgets.

Lifetime	Retained objects
Application	Shared services, configured stores and Core authorities.
Open Suite	Lifecycle presenter, membership/shell models, current live Snapshot/Notebook context and secondary windows.
One chart session	Dataset generation, attempts, Studies, presentation and camera. Independent even for identical markets.

Production versus legacy
The active Suite is in gui/research/. The older gui/windows/research_suite_window.py and gui/presenters/research_presenter.py are not the production composition. Their isolated tests do not establish current Suite behavior.

Authorities, state and identities

Truth or state family	Authoritative owner
Market identity / OHLCV validity	MarketId / canonical OHLCV store and validator
Research admission and full immutable data	AcceptedDatasetCatalog / HistoricalDatasetLoader
Disposable slices and camera	ResidentSliceService / HorizontalViewport
Chart membership versus placement	ResearchWorkspaceState / ResearchWorkspaceShellState
Generation, attempt acceptance, Studies	ChartSessionState; presenter sequences GUI operations
Mathematics / durable calculation identity	Financial Tools / Artifact services and Portable Recipe store
Research saved context	Study Setup, Snapshot and Notebook domains/stores
Saved Notebook assignments	Snapshot/Notebook link service and the Snapshot reference
Execution, terminal tasks, window tracking	CoreRunner + TaskManager / WindowRegistry + GuiWindowTracker

Identity is not appearance

Identity	Meaning and allowed change
Slot / session / generation	Slot 1..16 may be reused after removal. A new session has a new identity. Generation distinguishes source epochs within it.
Workspace position	Placement 1..16. Moving/swapping does not rename sessions or Studies. Detached positions remain reserved.
Study / Environment entry	Study ID is chart-local runtime identity. Saved entry IDs map to fresh Study IDs when applied.
Snapshot chart reference	Saved chart identity inside one Snapshot; restoration creates fresh runtime sessions.
Pool ID / name / content hash	Environment, Snapshot and Notebook IDs survive Update. Names are display metadata; hash changes with saved content.

A task ID retained by a presenter is correlation state, not a second task registry. A Manager list, disabled button or runtime summary is a projection of its owner's state, not another persistence or admission authority.

OHLCV admission and loading

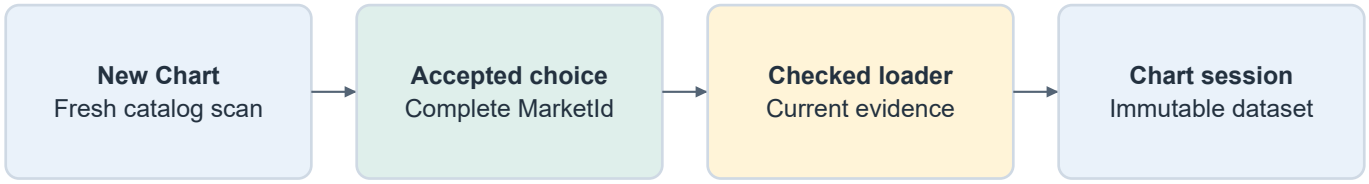


Figure 3. Selection does not waive the loader's later checks. The same market can be opened in multiple independent sessions.

Stage	Required result
Catalog	Canonical CSV and sidecar exist and agree on MarketId. Persistence is committed/repaired; validation is ok; rows > 0; current CSV hash matches accepted evidence and inspection is stable.
Selection	Exchange -> market type -> asset -> timeframe. Below 16 charts; current selection; lowest free logical slot plus a free position.
Full load	Strict CSV header and complete rows, increasing timestamps, finite OHLCV, valid high/low bounds, volume >= 0. Verify full hash, row count, endpoints and unchanged source/sidecar.
Publication	Matching task/session/generation only. A resident request prepares the visible slice. Failure shows an error; no partial or stale dataset is published.

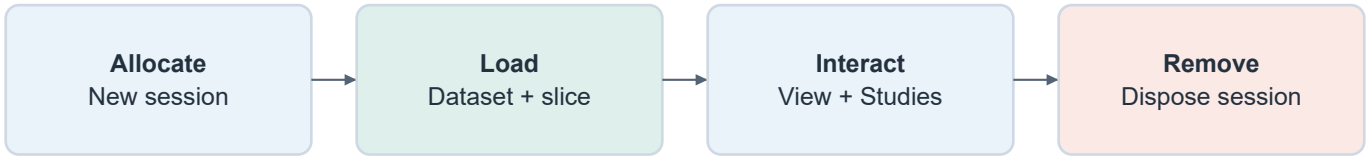
Full data and resident projection are different

The loader keeps up to four immutable full datasets by default and serializes loads per MarketId. A resident slice is a half-open range of the full axis; its cache key includes MarketId, fingerprint and range. Calculations use the full accepted axis, not the moving resident window.

Resident defaults	Meaning
2,000 visible + 1,500 each side	Target 5,000 real rows, shifted/clamped at boundaries; smaller sources stay smaller.
128 cached slices	Disposable LRU projections. New source fingerprint invalidates old cached slices for that market.
Nearest UTC timestamp	Binary search, earlier candle on equal distance, clamp beyond available coverage.

After OHLCV changes
An open chart keeps its original immutable dataset and Studies. A later selector opening rescans accepted data; a newly opened or restored chart loads the current accepted source. Live hot-reload is deferred.

Charts, panes and navigation



Action	Existing behavior
Multiple charts	Same-market charts can share immutable data while Studies, camera, styles and panes remain independent.
Move / swap	Change shell positions. Occupied attached positions swap. Detached chart positions cannot be moved into or moved while detached.
Detach / dock	Reparent the same panel/session. Closing the detached window docks it; explicit chart Close removes it. Detached charts still count toward 16.
Close chart	Retire presenter, attempts and chart-local dialogs; release slot/position. Other sessions retain identity; active fallback is the smallest surviving slot.
Pane structure	Price first; optional base volume is separate from a Volume Study. Oscillators have separate panes. Bottom eligible pane owns the shared time axis.

Camera and timestamp behavior

The camera includes 1,000 empty positions on each side by default. Visible count starts at 500, is limited to 20..2,000 within the domain, and initially aligns the last real candle to the right. Padding is view space, never synthetic OHLCV.

Interaction	Effect
Pan / zoom / refill	Move the camera; refill when real data approaches a resident edge. Existing full Study values are projected, never recalculated just because the camera moves.
Go To	Resolve a UTC timestamp to the nearest real candle, center and refill. The dialog is single-instance for that chart.
Pan Anchor	Broadcast user-originated horizontal timestamp intent to ready charts. Each resolves its own axis; initial load/restore/programmatic movement does not feed back.
Autoscale / styles / crosshair	Local view changes. Shared horizontal camera across panes; vertical price bounds remain chart-local. Dynamic overlays do not become calculated Studies.

Exact zoom, anchor, range, refill-threshold and edge-tie algorithms remain in D 10. No navigation or style action writes a saved pool object by itself.

Sixteen charts, preserved display rules

Capacity is 16 total charts, including detached charts. Fit 8 uses the same adaptive arrangement for each group of up to eight attached charts. More than eight attached charts creates two vertical pages; it does not introduce a new grid rule.

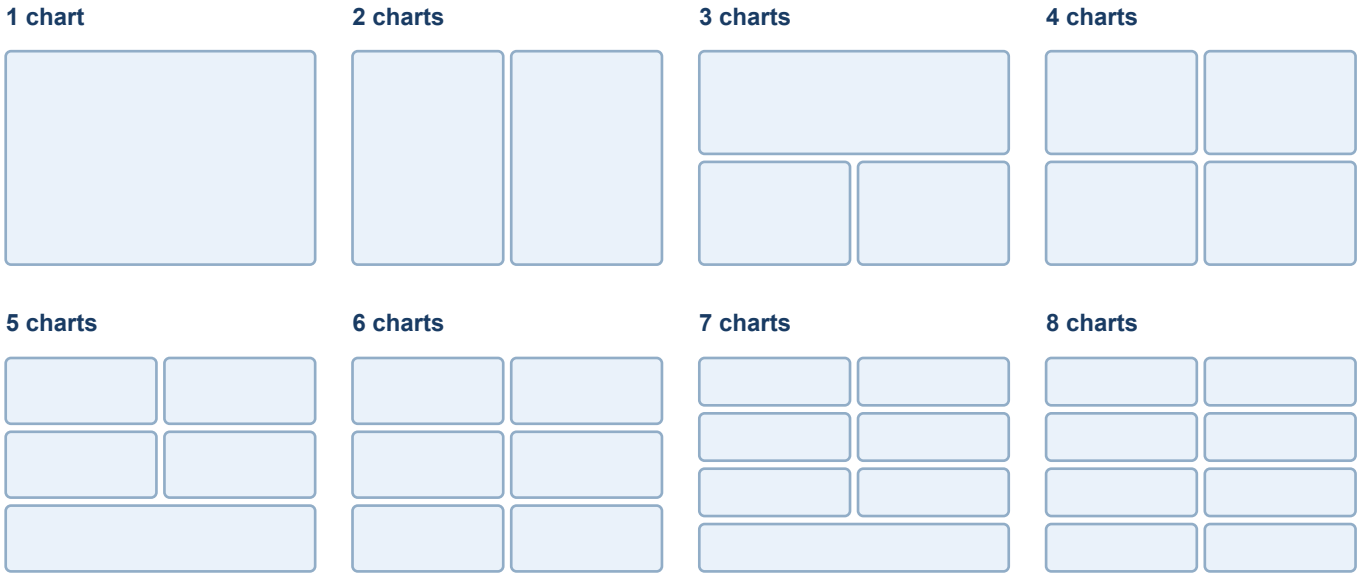


Figure 4. The original adaptive groups. Empty attached state has its own message; all-detached is a distinct empty-workspace state.

Attached charts	Fit 8 result
0..8	One adaptive group; no vertical scrollbar; minimum content height is one viewport.
9	Page 1: eight in four pairs. Page 2: one chart fills the page.
11	Page 1: eight. Page 2: three arranged full-width first, then one pair.
16	Two pages of eight, each four pairs. Scroll vertically between pages.

Each page has at least one viewport height. With two pages the content minimum is twice the viewport height plus the existing 2-pixel spacing. Compact panel density depends on that page having at least three rows. Order attached panels by position; holes do not renumber sessions.

Scroll 4 remains a separate mode

Use one scrolling adaptive layout. For four or more charts, place pairs in row order; an odd last chart from five upward spans both columns. Minimum content-height ratio is max(1, row_count / 2). Reparented panels stay visible through mode changes and detach/dock.

Layout changes alter presentation only. Snapshot cardinality accepts 1..16; no schema-version bump accompanied the capacity change.

Studies and Financial Tools

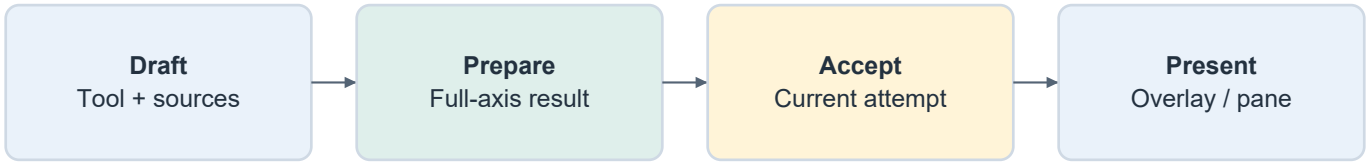


Figure 5. Apply produces a transient chart Study only after currentness and result validation. It does not save a Recipe, Artifact or Research pool object.

Operation	Mutation and refusal rules
Apply calculation	Resolve canonical parameters and eligible sources; calculate full accepted axis; verify tool, outputs, timeline, NaNs and dependencies before publishing.
Apply saved Artifact	Validate exact market, source lineage, tool/output semantics and full axis; load saved values without recalculation. Missing/stale lineage blocks application.
Edit	Block live dependents and self-dependency. Prepare replacement first. Success retains Study ID/name/metadata and compatible style, clears saved link, and becomes calculation mode. Failure keeps old result.
Remove / style	Removal requires no live dependents and deletes only the transient result. Style, visibility and pane changes affect presentation; saved counterparts are unchanged.
Multi-Artifact batch	Apply sequentially to the same current session. Stop on first failed/stale item; already applied items remain. This is not Environment rollback.

Sources: explicit OHLCV roles, same-session Study outputs, or exact saved Artifact outputs. Raw Construct sources are Open/High/Low/Close only. Raw Volume is prohibited; eligible calculated outputs follow the canonical source policy.

Research catalog	Tool keys
10 indicators	sma, ema, tema, hma, kama, bb, hck, strategy, peaks_troughs, universal_trend_classifier
7 oscillators	rsi, arsi, tdirsi, smi, mfi, obv, volume
8 constructs	derivative, angle, braids, braid_instability, delta, trap_area, percent_span_angle, angle_momentum

Research exposes 25 tools. Global dynamic_binning is excluded from Research calculation, Artifact and Environment application. Numerical formulas remain Financial Tools authority; style drivers and analysis usability are not interchangeable with renderability.

Saving results and DMS integration

Save consumes the existing complete result
Study Save does not rerun the calculation or persist only the visible slice. Dependent Study sources must already have valid durable links; Research does not silently save their dependency chain.

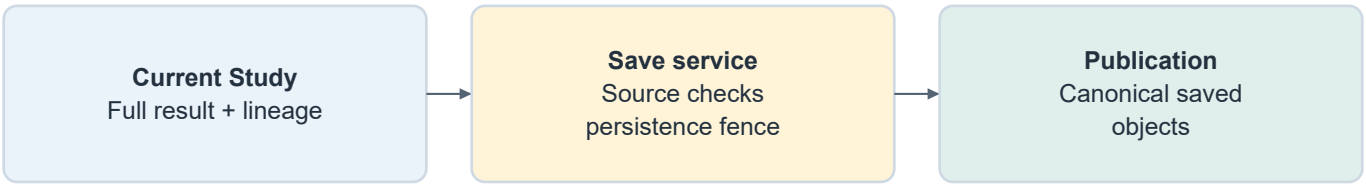


Figure 6. Explicit publication, then a saved link is accepted only by the still-current chart session.

Published concept	Scope and owner
ArtifactRecipeV1	Market-bound saved calculation recipe under canonical Artifact services.
Calculated / managed Artifact	Market- and source-specific values, metadata and lineage; managed version records use existing publication authorities.
PortableRecipeV1	Global reusable definition in the DMS recipe pool, separate from market-bound Artifact ownership.
StudySaveOutcome	Chart Study ID, saved link and created/reused result. Repeated Save validates and reuses the existing publication path where applicable.

Publication and failure boundaries

Resolve exact sources, recheck accepted OHLCV evidence, persist full values and verify the result, publish portable definitions and managed records, then return the outcome. Preserve full precision, categorical values, output order and NaN placement.

If publication fails, attempt cleanup of a newly created invalid/stale Artifact. Do not delete a pre-existing Artifact as cleanup. Report the original error and any cleanup failure. This path is not a promised crash-atomic transaction across all files.

The Financial Tools form's Save calculates/applies its draft once, then uses Study Save. If Apply succeeds but persistence fails, the resulting Study remains transient; it must not be presented as durably saved.

DMS integration stays explicit
DMS can discover saved publications. Research does not import them into a Database, subscribe a Database to future collection changes or modify DMS append/update rules. DMS Open Notebooks... reuses the same Research composition and Notebook Manager.

Study Environment lifecycle

An Environment stores a nonempty ordered set of Study definitions, metadata and supported presentation. It does not store candles, full numerical results, windows, tasks or a camera. Runtime Study dependencies become references to earlier saved entries.

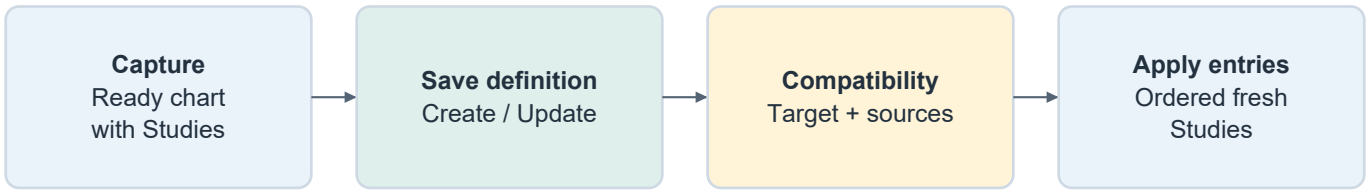


Figure 7. A pool definition and applied runtime Studies have independent lifetimes.

Operation	Behavior
Create / Update	Create requires a unique valid name. Update retains ID/creation time and can change name, description and definitions. Empty capture is refused.
Manage / Load	Manage can update name, description and entry metadata without applying. Load presents those details read-only. Target changes require current compatibility.
Calculation mode	Recalculate definitions on the target accepted full axis. A calculated Study remains calculation mode even if it was previously saved.
Artifact mode / sources	Any Artifact reference requires exact created_from MarketId and current lineage. Calculation-only Environments are reusable across accepted markets.
Append	Default: apply entries sequentially alongside existing Studies. Map each saved entry ID to its new runtime Study ID.
Replace	Build the complete new set first, then remove old Studies in reverse dependency order at finalization.
Delete	Remove the pool definition only. Applied Studies, Snapshot-embedded copies, Artifacts and Recipes survive.

Failure and cancellation

Normal calculation-stage failure, cancellation or stale target removes this run's additions in reverse order and preserves preceding Studies. Late delivery cannot restart a cancelled run. A retry is a new run.

Precise recovery limit
An unexpected failure while removing old Studies during Replace finalization is reported and the new set is cleaned up. Already removed old runtime Studies are not guaranteed to be resurrected. This is not a durable multi-object transaction.

Workspace Snapshot capture

A Workspace Snapshot is an explicit saved capture of one to sixteen ready charts and their workspace context. It is not autosave and does not freeze the original CSV bytes or calculated arrays.

Captured layer	Saved content
Workspace	Visualization mode (Scroll 4 / Fit 8), Pan Anchor and active chart reference.
Chart identity and location	Saved chart reference, full MarketId, unique position 1..16 and detached flag.
View state	Center timestamp and visible count; automatic/manual price scale; base-volume visibility; pane order and sizes.
Studies	Full embedded Environment definition or null. Runtime Study IDs become embedded entry IDs; oscillator pane references follow those IDs.
Notebook assignment	Optional Notebook ID only. Notebook content stays in its independent pool.
Object metadata	Snapshot ID, name, description, created/updated UTC timestamps, schema version and content hash.

Copies and references have different lifetimes

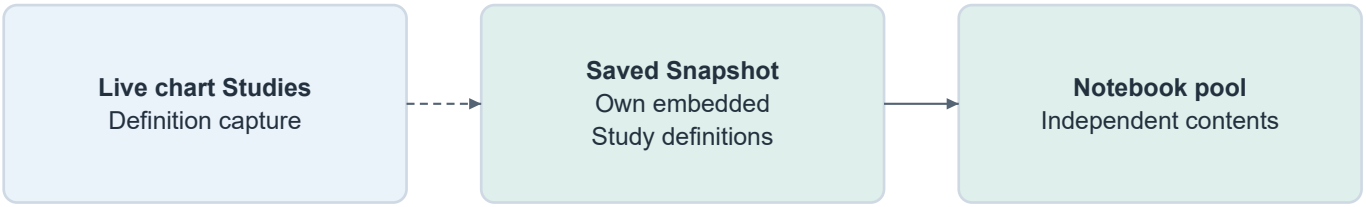


Figure 8. Left: the Snapshot embeds captured chart definitions, not a link to the Environment pool. Right: the Snapshot stores a Notebook reference, not an embedded copy.

Create allocates a new Snapshot ID and does not infer an assignment from the open editor. Update retains ID, creation time and saved assignment. The GUI keeps the Update name read-only and permits description changes; this GUI rule is not schema-level name immutability.

Capture and load have separate admission
Capture validates every chart and busy-state constraints. A structurally valid saved Snapshot may still fail restore preflight because data or Artifact lineage changed, a Notebook is absent, capacity is insufficient or work is busy.

Deleting a Snapshot removes its pool file. Current live identity/assignment is cleared if it was the active Snapshot; its charts, assigned Notebook, accepted sources, Artifact pools and standalone Environments survive.

Workspace restoration and recovery

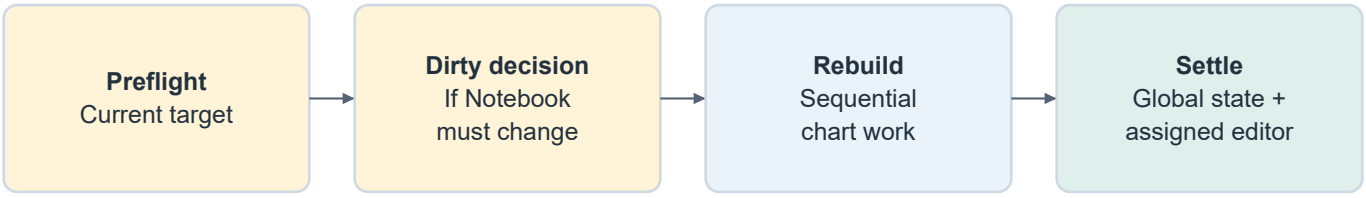


Figure 9. Dirty decisions precede teardown. Save requires fresh preflight and unchanged target/editor identity before continuing.

Phase	Required behavior
Preflight	Check idle state, accepted sources, embedded definitions, Notebook availability and capacity. Report global/per-chart blockers and warnings. Cancel before Load changes nothing.
Append position plan	Keep saved positions that are free; otherwise reserve the lowest free position in sequence. Detached reservations count. Never evict an existing chart.
Replace start	Capture rollback definition and previous live identities, then remove old charts. Append keeps existing sessions and records their global context.
Each chart	Allocate -> load current accepted dataset -> resident slice -> embedded Environment -> viewport/scale/volume/panes -> position/detach. Correlate every callback with the run.
Successful settlement	Both modes adopt the saved mode, Pan Anchor, active chart, Snapshot ID and Notebook assignment. Retire stale dialogs/New Chart intent; retain or open the assigned editor.

Notebook operations cannot compete with pending/active restore. Manager refresh is deferred; a late-created Manager is busy. Success retires the old Manager and drops its refresh. On refusal/failure a surviving current Manager refreshes once; a closed or replaced Manager stays closed.

Failure	Recovery
Before teardown	Keep live charts and editor. A successful explicit Notebook Save remains saved even if fresh preflight later refuses.
Append after additions	Remove only the run's added charts; restore preceding global context and live identities.
Replace after teardown	Reconstruct the captured workspace through normal load/apply paths. Fresh sessions and current accepted data replace the old objects.
Rollback also fails	Report both failures explicitly. Never claim atomic recovery or resurrection of old dataset bytes/Qt objects.

Active restore/rollback disables Cancel and rejects close/Escape until terminal settlement. This composite is not advertised as user-cancellable. The outgoing editor buffer survives failed restore, including a prior Discard whose transition did not succeed.

Notebooks, notes and annotations

A Notebook is an independent saved research object, with zero to 64 unique MarketId pages. One editor holds a draft; chart annotations derive from its last valid draft. A Notebook can exist without charts or Snapshot assignments.

Row family	Content and meaning
Note	Stable row ID, optional nonnegative UTC timestamp, nonempty text. Untimed Notes are allowed. Plain Notes are not chart annotations.
Potential Trade	Timestamp, long/short direction, optional positive finite entry/target/stop prices, status, outcome and note. These are observations, not broker orders or automatic trade calculations.
Point of Interest	Timestamp, optional positive finite price, nonempty title and optional description. Represents a marked observation on the market page.

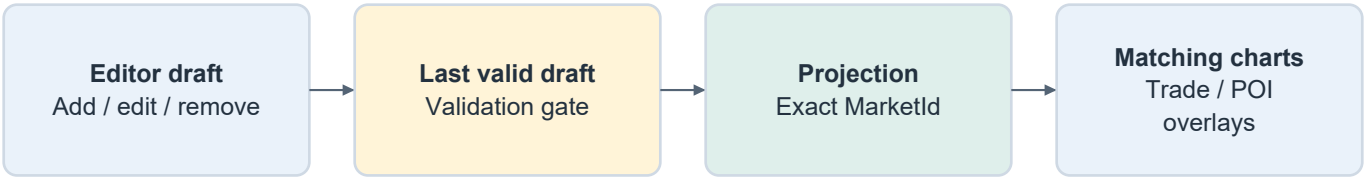


Figure 10. Preview changes presentation only; an invalid intermediate draft cannot publish a malformed Notebook or change Study results.

Creation, saving and navigation

Create opens a dirty new draft with pages for current chart markets. Pages may also be added from the current chart. Save updates the existing Notebook ID; Save As creates a different pool object and does not rewire Snapshots that reference the original.

Rows have IDs unique across the whole Notebook. Pages use complete MarketId, including timeframe. Projection never spreads notes from one timeframe into another. Trades use L/S labels; Points use +. If price is absent the chart supplies its anchor placement.

The Notebook controls annotation visibility and pixel offsets. Closing the editor removes its annotation projections, not saved content. Preview is suspended during restore.

Go To from a timestamped row	Result
Active chart matches	Navigate that chart through the normal timestamp/resident path.
Another chart matches	Use the first matching chart by workspace position.
No match / restore active	Report no matching chart or refuse during restore. Do not silently create another market chart.

Trade status: planned/open/closed/cancelled. Outcome: pending/win/loss/breakeven. These values are user records, not a trading lifecycle implemented by Research.

Dirty changes and pending transitions

Ask before a destructive transition
Editor replacement/close, Suite or main-window close, Clear, and a restore that must change Notebook pass through the existing dirty guard. Keeping the same Notebook during restore adds no dirty prompt.

Editor state / decision	Continuation and durable result
Clean	Proceed only if other admission guards also allow the operation.
Dirty + Cancel	Refuse the entire requested transition. Keep dirty editor, charts and relevant windows; saved content is unchanged.
Dirty + Discard	Proceed without saving. For restore, keep outgoing buffer until restore succeeds, so a failed restore can preserve it.
Dirty + Save	Validate and submit one explicit save. Retain the intended continuation until success; then recheck the transition's current guards.
Invalid / failed Save	Report reason, clear pending continuation and retain editor. Do not assume success or tear down charts.
Save already pending	Keep one compatible continuation; reject conflicting requests. Do not submit duplicate saves.
Other Notebook/link work	Refuse conflicting transitions. Do not create a parallel Notebook mutation path.

Two composed outcomes matter

Workflow	Correct ordering
Save -> restore	Finish Save; rerun preflight; confirm editor/session/position/dialog identity; reserve Notebook work from Manager refresh; restore charts and open assigned editor.
Save -> Clear	Run the existing Clear continuation before Manager listing. Accepted Clear retires Manager, so no refresh is needed.
Independent conflict after Save	The existing guard may refuse Clear. Save remains durable; editor may already be closed; charts survive; current Manager refreshes. No automatic retry.

Deletion is a different decision
If Manager is deleting the same Notebook that is open and dirty, only explicit Discard permits that deletion path. Save or Cancel stops deletion; the implementation does not Save and then silently delete the Notebook.

Accepted closure, rather than a mere close request, controls window tracking. Cancel must leave the Suite/main window registered as open with no closed timestamp. A disabled button is only a view aid: current controller admission is checked again.

Assignments and Notebook deletion

A Snapshot refers to zero or one Notebook. Multiple Snapshots may refer to the same Notebook. The link is saved in each Snapshot; a Notebook neither belongs to a Snapshot folder nor owns a reciprocal list.

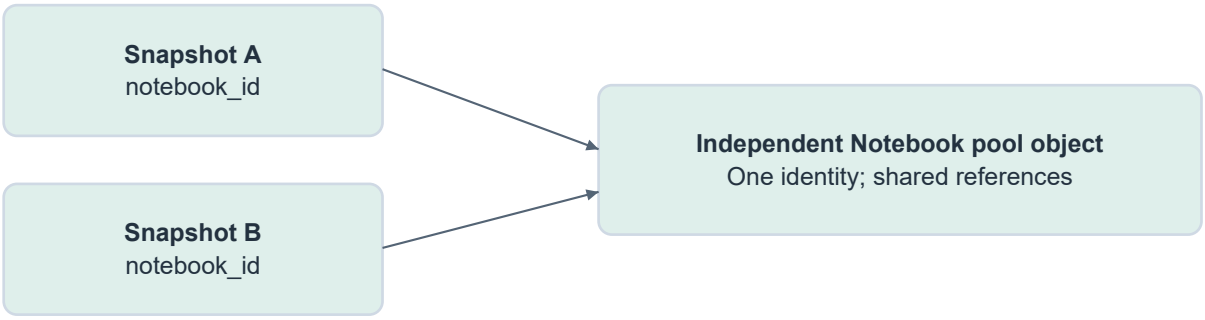


Figure 11. References do not transfer ownership. Unassign or delete a Snapshot and the Notebook survives.

Operation	Contract
Assign / reassign	Validate both objects under the link-service lock. Change the Snapshot reference; preserve the previous Notebook. Assigning the same current ID is idempotent.
Unassign	Require the expected current Notebook ID; a stale mismatch fails. Clear only the reference; preserve Notebook content.
Update live indicators	After a successful mutation of the live Snapshot, synchronize its live assignment display. Resetting a live indicator alone is not a saved unassign.

Coordinated deletion sequence

1. Confirm Notebook deletion. If that editor is dirty, only Discard allows this path; close the editor before submitting delete.

2. In one coordinated fenced operation, load the Notebook and enumerate valid saved Snapshot references.

3. Clear referring Snapshot links, then delete the Notebook file.

4. If cleanup/deletion fails, attempt reference restoration in reverse order. Report primary and rollback failures honestly.

Recovery restores references, not all historical bytes
Restoring a link may advance Snapshot timestamps and content hashes. Invalid/unreadable Snapshot summaries are not silently repaired by the scan. No chart definitions, OHLCV, Studies or Artifacts are deletion targets.

Background work and stale results

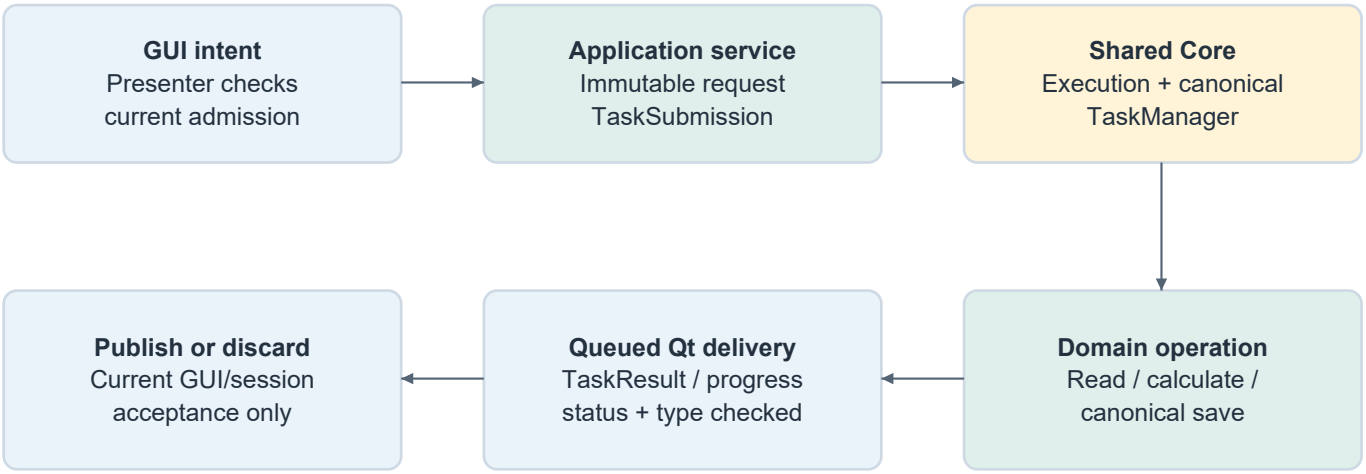


Figure 12. Worker code does not touch Qt widgets. Submission failure and failed/cancelled terminal delivery are distinct paths, both releasing matching pending state.

Boundary	Acceptance rule
GUI target	Presenter is not disposed; retained task/run/dialog identity still matches. Reopened Suite or replaced dialog cannot receive an old callback.
Chart target	Exact session, generation, request, MarketId and source fingerprint where relevant. Slot number alone is insufficient after reuse.
Study target	Expected full axis/output schema and current dependencies. Prepared result is not live state until accepted.
Progress	Present matching progress only. Research Activity and Manager summaries are read models, not alternative task state.

Cancellation must describe what actually happened

Read/calculation cancellation is cooperative and may prevent publication. Durable operations use their existing persistence fence: after it is crossed, cancellation may be refused and a save may finish after the window closes. Core retains the real terminal result; retired GUI delivery is dropped.

Pan, zoom, style, focus and detach/dock remain direct GUI work. Composite restores and Environment runs correlate individual Core tasks; there is no parent-child umbrella task hierarchy. Research launches no child operating-system process.

Operation names and all tracked window IDs are listed in S 22-23. Cancellation bookkeeping is removed at terminal completion even when its GUI owner no longer exists.

Clear, close, reopen and shutdown

Action	Accepted transition	What survives
Clear Research Suite	Check busy guards -> confirmation -> dirty decision -> retire charts/editor/dialogs and pending selector intent; reset live IDs, Pan Anchor off, Scroll 4 on, Activity reset. Suite remains open.	All saved pools, accepted OHLCV and saved assignments.
Close Suite	Dirty guard first; invalidate intents and GUI delivery; cancel cancellable work; dispose sessions/dialogs; composition forgets the pair.	Shared application services, Core and durable objects.
Reopen Suite	Build a fresh Suite with fresh sessions/dialogs. Empty workspace, Scroll 4, Pan Anchor off; no automatic saved-object load.	Previously saved pools remain available for explicit load.
Main close / Exit	Research dirty guard runs before main window or other suites are closed. Cancel preserves them all. Accepted shutdown belongs to Leonardo.	Saved state; Core records actual task settlement.

Clear is a live-state operation

Confirmation No changes nothing, including a dirty editor. Dirty Cancel aborts the whole Clear. A busy conflict refuses the operation without bypassing a guard or queuing an automatic retry. A successful Notebook Save happens before teardown.

Closing is not cancellation of history
A durable save already past its fence can settle after GUI retirement. Research does not kill worker threads or fabricate a completed/cancelled state to make shutdown look finished.

Window identity and accepted close

Tracked windows are single-instance for their function and context. Repeated opening focuses/reuses the retained window. A different chart or Study has a different context ID; the title is not its identity. Short confirmation message boxes are not separate durable runtime windows.

GuiWindowTracker and WindowRegistry must record closure only after it is accepted. A dirty Cancel preserves the actual window, registry open state and unsaved editor. Late progress/results cannot update a newly opened Suite.

Protected roots remain untouched by Clear/close/restore: historical_data/, data_manager/, study_environments/, workspace_snapshots/ and research_notebooks/. No cleanup or normalization is implied by those actions.

Persistence and compatibility

Pool and canonical file	Saved format	Contents
study_environments/<id>.json	1.0 study_environment	Ordered definitions and supported presentation; no numerical arrays.
workspace_snapshots/<id>.json	1.1; reads 1.0 research_workspace_snapshot	1..16 chart captures, embedded definitions, view state, optional Notebook reference.
research_notebooks/<id>.json	1.0 research_notebook	0..64 MarketId pages; Notes, Potential Trades, Points of Interest and annotation settings.

Shared store contract

1. Use configured contained paths; reject unsafe paths, symlinks and junctions. Lock store mutations. Create refuses existing ID or case-folded duplicate valid name.

2. Update keeps object identity and creation time; updates content, timestamp and hash. Lists expose invalid summaries with reasons; they do not repair files.

3. Serialize UTF-8 with sorted keys, compact separators, ensure_ascii=False and allow_nan=False. Hash the payload without content_hash and without a trailing newline.

4. Environment bytes have no trailing newline. Snapshot and Notebook canonical bytes have one. Exact load verifies canonical bytes, schema, identity and hash.

5. Write a same-root unique temporary file, flush and fsync, atomically replace the destination; clean the temporary file on failure. This is per-file replacement, not a cross-pool crash transaction.

Readable JSON is not permission to hand-edit saved data
Pretty-printing a canonical stored file can make it fail exact loading. The complete field lists and parser-verified readable examples belong to the data-contract manual; use canonical serializers for actual store bytes.

Compatibility distinction	Meaning
Snapshot 1.0 versus 1.1	1.0 omits notebook_id and cannot carry an assignment. Reading does not rewrite it; newly built/updated Snapshots use 1.1.
Sixteen-chart capacity	No format-version bump. Older builds limited to eight cannot load a larger Snapshot.
Format versus current sources	Valid structure/hash does not prove source availability or lineage. Restore/application still requires current compatibility and load checks.

Survival rules and a complete workflow

Operation	Removed or changed	Must survive
Close chart	Session and transient Studies/views	Saved results, definitions and observations
Remove Study	One dependency-safe runtime result	Artifact/Recipe counterpart and sources
Delete Environment	Standalone pool definition	Applied Studies and embedded Snapshot copies
Delete Snapshot	Pool capture; live ID if current	Live charts, Notebook, data and other pools
Unassign Notebook	Snapshot reference only	Independent Notebook contents
Delete Notebook	Notebook + coordinated valid references	Snapshot chart definitions, Studies, Artifacts, data
Clear / close Suite	Live research context	All saved pools and saved assignments

Follow one investigation through its full lifecycle

1. Open two charts for accepted BTCUSDT 1m. Apply an SMA and an eligible dependent Construct on the first. The second chart remains unchanged.

2. Save required source Studies before saving the dependent result. DMS discovers the canonical publications; no Database membership changes automatically.

3. Save the first chart's definitions as an Environment. Save both charts and view settings as a Workspace Snapshot.

4. Create a Notebook page for that exact market/timeframe; record observations and save. Explicitly assign the Notebook to the Snapshot.

5. Clear the Suite. Live charts/editor disappear; all saved objects and the Snapshot's saved assignment remain.

6. Restore the Snapshot. Current accepted data load; calculation definitions recalculate; Artifact-mode entries require exact current lineage. Views restore and the assigned Notebook opens.

7. Delete the standalone Environment: the Snapshot keeps its embedded copy. Unassign the Notebook: its contents remain in its pool.

Completion is the whole composed outcome
Charts restored but the assigned editor missing is an incomplete restore. A successful Save with residual charts after an otherwise admissible Clear is an incomplete Clear. A successful callback alone is not the lifecycle contract.

Evidence, limitations and stopping point

The map describes the active implementation after the final A4 Notebook transition correction and the accepted 16-chart extension. It introduces no new behavior or architecture proposal. The detailed manuals remain the implementation reference.

Evidence already recorded	Result and scope
A4 full repository validation	2,831 passed; 7 skipped. This run includes the final patch and follows the DMS Open Notebooks button change.
A4 focused / Research / native	328 focused passes; 1,351 Research passes with 5 skips; 127 native checks. Overlapping selections, not additive unique coverage.
Independent A4 audit	115 offscreen checks and 21 native checks on the unchanged candidate. Both Notebook ordering findings closed; sixteen-chart rules preserved.
Detailed-document validation	21 local links/headings, 16 persisted model classes, 3 complete canonical payload round trips, 31 cited test files and 34 layout cases checked.

These are prior code/document-validation receipts, not tests newly run while making this PDF. The seven full-suite skips reflect Windows symlink privilege. Existing warnings and an initial corrected external harness assertion are retained in the audit record.

Present limits and deferred work

Category	Boundary
Known recovery limits	Replace rollback can fail; it reloads current accepted data into fresh sessions. Multi-file publication/link cleanup is not a crash-atomic transaction.
Deliberate runtime limits	Cooperative cancellation; fenced writes may finish after close. No parent-child umbrella Core task for composite Research workflows.
Deferred / separate	Live OHLCV hot-reload, Dynamic Binning in Research, application Appearance editor, arbitrary redesign, Analysis/Trading workflows.
Settled DMS boundaries	Original intermittent Windows CSV replacement cause remains unidentified. Compression, storage-format changes and large-file parallel loading remain deferred.
Acceptance distinction	Automated and native evidence supports the audited candidate. The user's personal visual/product acceptance remains separate.

Reconstruct Research within Leonardo by preserving canonical external contracts and the observable acceptance matrix. Reimplementing the entire Financial Tools library also requires its own specifications and fixtures; this map does not create competing mathematical definitions.

No additional product correction is proposed by this documentation task. The stop condition is a coherent, traceable companion map alongside the existing full documents.

Implementation and source map

Responsibility	Primary current modules
Composition and windows	core/app.py; core/config.py; gui/composition.py; gui/research/composition.py; gui/window_tracking.py
Production Suite and charts	gui/research/suite_window.py; lifecycle_presenter.py; gui/presenters/research_chart_presenter.py; gui/research/chart_panel.py; workspace_widget.py
Membership / placement / camera	research/workspace.py; workspace_shell.py; session.py; viewport.py; gui/widgets/research_workspace_layout.py
Admission and immutable data	research/catalog.py; dataset.py; resident.py; application.py
Studies and presentation	research/studies.py; study_execution.py; study_application.py; study_projection.py; study_presentation.py; gui/chart/
Tool setup and Environments	research/study_setup.py; study_setup_service.py; study_setup_application.py; study_environment.py; study_environment_store.py
Snapshots	research/workspace_snapshot.py; workspace_snapshot_service.py; workspace_snapshot_application.py; workspace_snapshot_store.py
Notebooks and references	research/notebook.py; notebook_service.py; notebook_application.py; notebook_store.py; workspace_notebook_link.py
Canonical financial boundary	financial_tools/specifications.py; calculation_models.py; calculation.py; construct_input_eligibility.json

After a folder prefix, adjacent short filenames in that cell use the same folder. Full paths, ownership detail and external Core/DMS references are in A 6-7. Older similarly named GUI classes are not the production entry point.

Reconstruction sequence

1. Establish external contracts and configured injection. **2.** Implement validated immutable models and canonical stores. **3.** Prove admission, loading, slices, attempts and camera. **4.** Implement Study Apply/Edit/Save and lineage. **5.** Build chart/panes, membership, shell and layout. **6.** Add Environment application and Snapshot recovery. **7.** Add Notebook drafts, assignments and dirty transitions. **8.** Wire actual composition, DMS entry, tracking and shutdown.

Where to go next in the document set

Scope and commands: S. **Dependencies and authority:** A. **Ordered transitions, failures and surviving references:** L. **All persisted fields, exact algorithms, complete JSON examples and executable acceptance criteria:** D.

One reference set, two reading depths
The map explains how the pieces fit together. The manuals specify their complete behavior and contracts. Keep both synchronized when an approved behavioral change is made; the map never replaces the manuals.